

CIFAR-10 CNN Experiment Report

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1. Data Augmentation

1.1 Additional augmentation layers

Three augmentation layers were tested individually while keeping the rest of the CNN unchanged.

RandomContrast

Training accuracy: 0.4345

Validation accuracy: 0.4408

RandomContrast produced the best result among the augmentation variants. The increase in validation accuracy suggests that moderate contrast variation improved generalization without removing useful image structure.

RandomBrightness

Training accuracy: 0.1005

Validation accuracy: 0.1000

RandomBrightness performed at chance level. This indicates that the transformation was too disruptive for CIFAR-10 images and prevented effective learning.

RandomTranslation

Training accuracy: 0.3940

Validation accuracy: 0.4341

RandomTranslation also improved validation accuracy relative to the baseline. This suggests that limited spatial shifts helped the model become less sensitive to object position.

Baseline Performance without modifying the augmentation block:

Final training accuracy: 0.4334

Final validation accuracy: 0.4539

Summary

Best augmentation: RandomContrast

Highest validation accuracy: 0.4408

1.2 Removing data augmentation

No augmentation

Training accuracy: 0.3953

Validation accuracy: 0.4049

Removing augmentation reduced validation accuracy relative to the augmented runs. The training-validation gap remained small, so there was no strong sign of severe overfitting in this short run, but overall generalization worsened.

Summary

Removing augmentation reduced test performance

Augmentation improved generalization

2. CNN Architecture

2.1 Increasing the number of filters

The original filter sizes (32, 64, 128) were increased to (64, 128, 256).

Training accuracy: 0.3964

Validation accuracy: 0.3183

Training time: approximately 200s and 197s

Increasing the number of filters made training much slower and reduced validation accuracy. Under the current setup, the larger model did not improve performance.

Summary

More filters increased cost

Validation performance decreased

2.2 Changing filter size from (3, 3) to (5, 5)

Training accuracy: 0.4153

Validation accuracy: 0.4901

Training time: approximately 258s and 256s

Using larger filters improved validation accuracy, suggesting that broader receptive fields captured more useful spatial features. However, training became substantially slower.

Summary

(5, 5) filters improved accuracy

Computational cost increased significantly

3. Pooling Layers

3.1 Replacing MaxPooling2D with AveragePooling2D

Training accuracy: 0.4408

Validation accuracy: 0.5022

Training time: approximately 69s and 67s

Replacing max pooling with average pooling improved validation accuracy and slightly reduced training time. This suggests that average pooling preserved more useful information for this task.

Summary

AveragePooling2D outperformed MaxPooling2D

3.2 Removing one pooling layer

Training accuracy: 0.4217

Validation accuracy: 0.5045

Training time: approximately 74s and 81s

Removing one pooling layer further improved validation accuracy. This suggests that the original architecture may have reduced spatial information too aggressively.

Summary

Removing one pooling layer improved performance

Preserving more spatial detail was beneficial

4. Regularization

4.1 Modifying dropout

Dropout = 0.3

Training accuracy: 0.4820

Validation accuracy: 0.5327

Training time: approximately 80s and 75s

Reducing dropout to 0.3 improved both training and validation accuracy. The model retained enough regularization while learning more effectively.

Dropout = 0.7

Training accuracy: 0.3240

Validation accuracy: 0.3252

Training time: approximately 77s and 81s

Increasing dropout to 0.7 reduced both training and validation accuracy. This indicates underfitting caused by excessive regularization.

Summary

Best dropout rate tested: 0.3

Very high dropout caused underfitting

4.2 Adding L2 regularization

Training accuracy: 0.4426

Validation accuracy: 0.4266

Training time: approximately 78s and 80s

Adding L2 regularization reduced validation accuracy in this run. The regularization likely constrained the model too strongly under the current settings.

Summary

L2 regularization did not improve validation performance

5. Optimizer and Learning Rate

5.1 Comparing optimizers

SGD without momentum

Training accuracy: 0.4926

Validation accuracy: 0.5041

Training time: approximately 77s and 74s

SGD with momentum

Training accuracy: 0.4973

Validation accuracy: 0.5447

Training time: approximately 76s and 83s

RMSprop

Training accuracy: 0.5247

Validation accuracy: 0.4731

Training time: approximately 77s and 73s

Among the tested optimizers, SGD with momentum achieved the highest validation accuracy. RMSprop reached the highest training accuracy, but generalization was weaker.

Summary

Best optimizer: SGD with momentum

Highest validation accuracy: 0.5447

5.2 Learning rate experiments

Adam, learning rate = 0.00001

Training accuracy: 0.1000

Validation accuracy: 0.1000

Training time: approximately 76s and 71s

This learning rate was too low. The model remained at chance level after two epochs.

Adam, learning rate = 0.001

Training accuracy: 0.0977

Validation accuracy: 0.1000

Training time: approximately 75s and 73s

This run also remained at chance level and failed to converge meaningfully.

Adam, learning rate = 0.1

Training accuracy: 0.1022

Validation accuracy: 0.1000

Training time: approximately 77s and 73s

This learning rate was too high. Training was unstable and failed.

Summary

Very low learning rates slowed learning too much

Very high learning rates caused training failure

6. Training Parameters

6.1 Increasing epochs from 2 to 10

Training accuracy: 0.5669

Validation accuracy: 0.5914

Training time: approximately 72 – 83s per epoch

Increasing the number of epochs improved both training and validation accuracy substantially. No strong overfitting was observed by epoch 10.

Summary

More epochs improved performance significantly

6.2 Changing batch size

Batch size = 32

Training accuracy: 0.4219

Validation accuracy: 0.4761

Training time: approximately 89s and 83s

A smaller batch size improved validation accuracy but increased training time.

Batch size = 128

Training accuracy: 0.4226

Validation accuracy: 0.4440

Training time: approximately 77s and 72s

A larger batch size reduced training time per epoch but lowered validation accuracy.

Summary

Smaller batch sizes improved generalization

Larger batch sizes improved speed

7. Visualization and Analysis

7.1 Training and validation curves

Training accuracy increased steadily and training loss decreased across epochs. Validation accuracy also improved for most of the training. After about epoch 7 – 8, validation accuracy began to flatten and decline slightly, while validation loss increased.

Summary

The model showed mild overfitting in later epochs

Early epochs produced the strongest generalization gains

7.2 Misclassified images

The misclassified examples were typically small, blurry, or visually similar to another class. Example errors included:

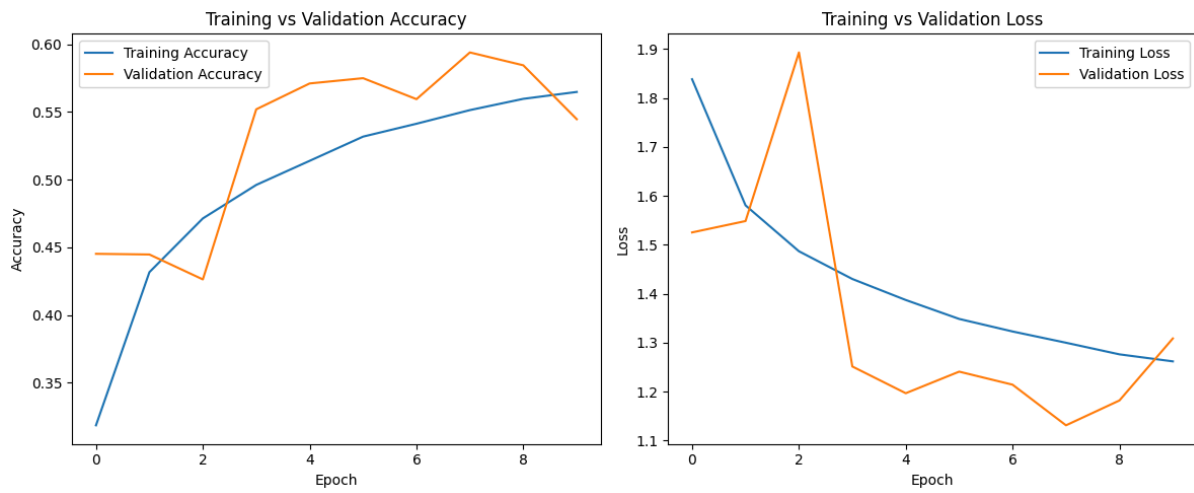
dog → bird

cat → automobile

Likely causes include low image resolution, weak object boundaries, distracting backgrounds, and visual similarity between classes.

Summary

Most errors were linked to low resolution and class similarity



8. Prediction Analysis

8.1 Precision and recall for the class cat

Precision: 0.4751

Recall: 0.1810

Precision for the cat class was moderate, indicating that predicted cat labels were correct about half the time. Recall was low, meaning that many true cat images were missed.

Summary

The model was conservative when predicting cat

Precision was moderate, but recall was poor

8.2 Confusion matrix analysis

Most frequently confused class pairs:

truck → automobile: 458

airplane → ship: 272

cat → frog: 253

deer → frog: 205

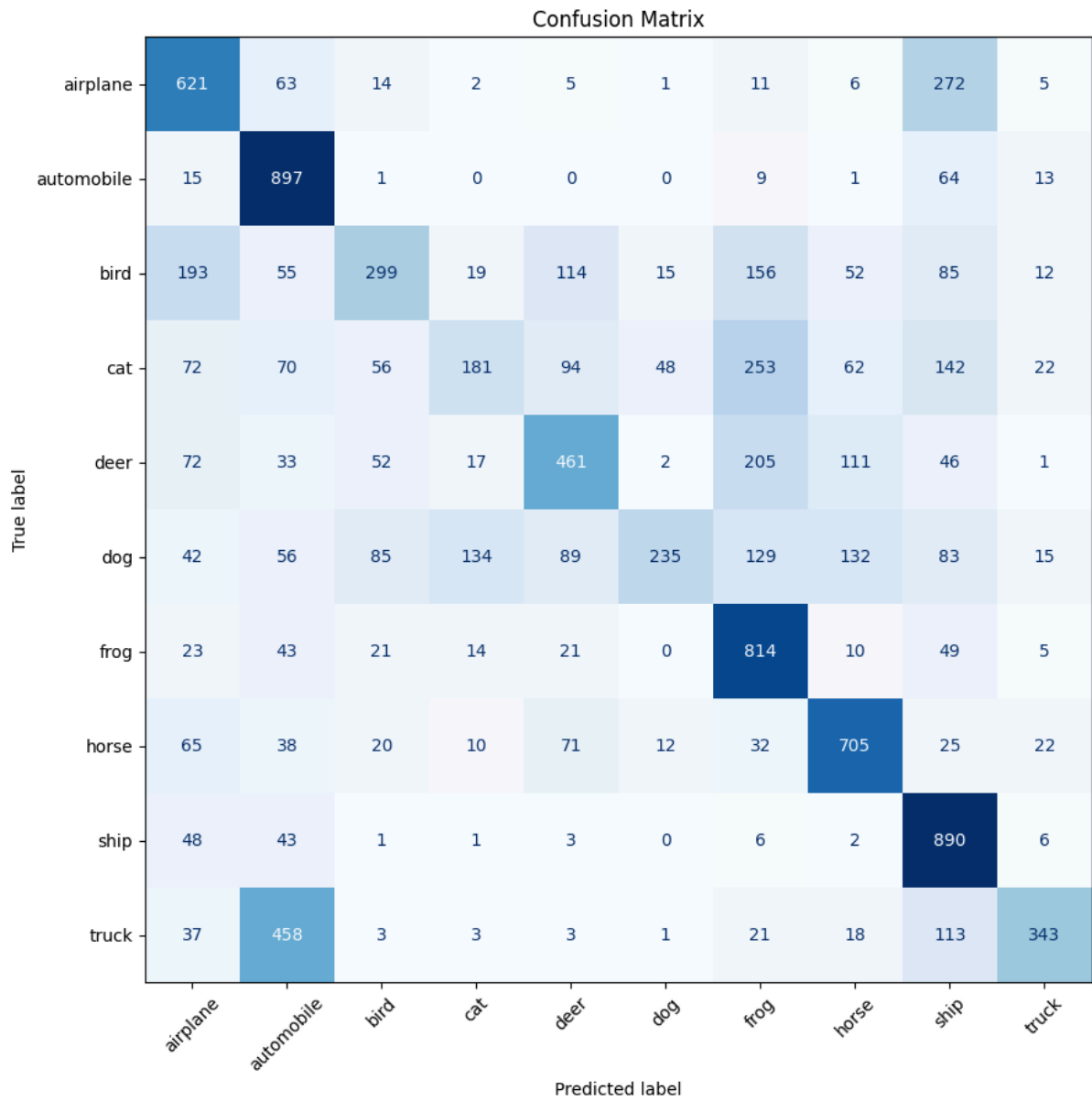
bird → airplane: 193

The confusion matrix shows that the model struggled most with visually similar classes. Vehicle classes were frequently mixed, and several animal classes were also confused, likely due to the limited resolution of CIFAR-10 images.

Summary

The largest errors occurred between visually similar classes

Limited image detail was a major source of confusion



Across the experiments, the strongest improvements came from:

1. RandomContrast augmentation
2. (5, 5) filters
3. AveragePooling2D
4. removing one pooling layer
5. dropout rate 0.3
6. SGD with momentum
7. increasing the number of epochs to 10

The results also showed clear tradeoffs:

- A. larger models and larger filters improved some results but increased training time
- B. smaller batch sizes improved validation accuracy but slowed training
- C. excessive regularization and poor learning-rate choices reduced performance

The best results were obtained when the model balance favored stronger feature extraction and moderate regularization without making optimization unstable.